A thermal-MPO derivation of the spin-1 gap bound

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Assuming two finite-size estimates from *The periodic spin-one Haldane gap*, we prove $\Delta_L > (\log 20)/784$ for every even $L \geq 2304$. An exact thermal matrix product operator (MPO) relates the purities of its spatial transfer operator and the Gibbs state. These purities control the excited-state weight as the chain grows. Numerical certification of the starting estimates is assumed.

We consider the antiferromagnetic spin-1 Heisenberg chain with periodic boundary conditions $\boldsymbol{S}_{L+1} = \boldsymbol{S}_1$ and $J = k_B = 1$:

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{1}$$

All system sizes are even and at least four. Let $E_0(L) \leq E_1(L) \leq \cdots$ include multiplicities and define the gap $\Delta_L = E_1(L) - E_0(L)$. For the Gibbs weights $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$,

$$\frac{w_1}{w_0} = e^{-\beta\Delta_L} \leq e^{-c\beta} \quad \Longrightarrow \quad \Delta_L \geq c. \tag{2}$$

To make c independent of L, we first use a spatial transfer operator to propagate a Gibbs-purity bound when L and β double. We then extend the bound to intermediate lengths and apply Eq. (2). In the diagrams, horizontal legs are virtual indices, vertical legs are physical indices, and connected legs are contracted [2, 3].

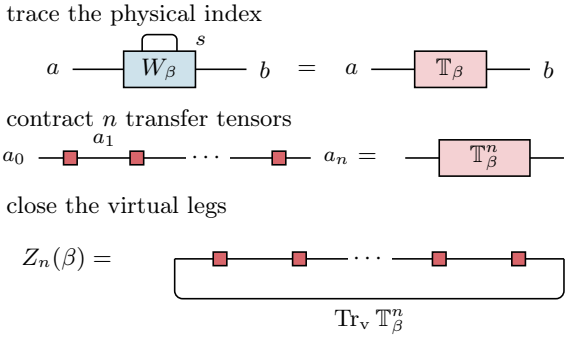
Thermal MPO.— To obtain a real symmetric transfer operator, use the Cartesian spin basis $s, s' \in \{x, y, z\}$. The matrices $(S^\mu)_{ss'} = -i\epsilon_{\mu ss'}$ give real operators $G_\mu = iS^\mu$, $\|G_\mu\| = 1$, and $-H_L = \sum_{j,\mu} G_{\mu,j} G_{\mu,j+1}$. Each bond event (τ, μ) in the exponential expansion inserts G_μ at both endpoints. Let a, b record these events on the two bonds adjacent to a site, with $0 < \tau < \beta$. The local MPO tensor is

$$W_\beta^{ss'}(a,b) = \left\langle s \left| \mathcal{T}_\tau \prod_{(\tau,\mu)\in a\cup b} G_\mu \right| s' \right\rangle. \tag{3}$$

Here $\mathcal{T}_\tau$ places later times to the left. Summing event labels and integrating ordered times defines a measure μ_β of mass $e^{3\beta}$, with unit mass for the empty history. Contracting L tensors around the ring recovers the absolutely convergent exponential series. This MPO is exact, with virtual space $L^2(\mu_\beta)$ and no bond truncation; for Trotter-based constructions, see Ref. [3].

The transfer kernel $\mathbb{T}_\beta(a,b) = \sum_s W_\beta^{ss}(a,b)$ is real, symmetric, and bounded in absolute value by three. It is therefore self-adjoint and Hilbert–Schmidt on $L^2(\mu_\beta)$. Its eigenvalues $\lambda_i(\beta)$ give, for even L,

$$Z_L(\beta) = \operatorname{Tr}_{\mathrm{v}} \mathbb{T}_\beta^L = \sum_i |\lambda_i(\beta)|^L. \tag{4}$$



The loop on W_β sums $s = s'$. The red chain has open virtual legs; closing them takes the virtual trace $\operatorname{Tr}_{\mathrm{v}}$ over $a_0 = a_n$.

Purity and excited-state weight.— We want one Gibbs weight to be close to one. For a positive trace-one operator with eigenvalues p_i and $p_{\max} = \max_i p_i$, the purity obeys

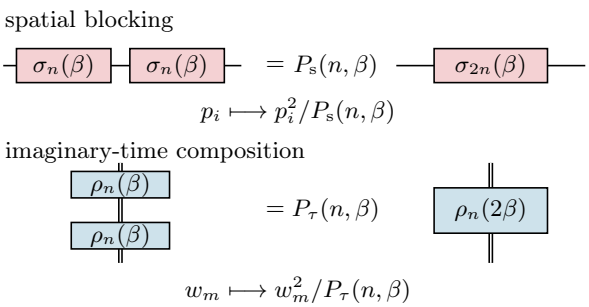
$$P = \sum_i p_i^2 \leq p_{\max} \sum_i p_i = p_{\max}. \tag{5}$$

Thus $1 - p_{\max} \leq 1 - P$. For the Gibbs state, $P > 1/2$ implies a unique ground state, with excited-state population $1 - w_0 \leq 1 - P$.

To propagate this bound, normalize both the transfer and Gibbs operators:

$$\sigma_n(\beta) = \frac{\mathbb{T}_\beta^n}{Z_n(\beta)}, \qquad \rho_n(\beta) = \frac{e^{-\beta H_n}}{Z_n(\beta)}. \tag{6}$$

Here σ_n is the open red chain divided by its closed-loop value. Both operators are positive with unit trace; denote their eigenvalues by p_i and w_m, respectively. Their purities are $P_{\mathrm{s}}(n,\beta) = \operatorname{Tr}_{\mathrm{v}} \sigma_n(\beta)^2$ and $P_\tau(n,\beta) = \operatorname{Tr} \rho_n(\beta)^2$. Squaring σ_n doubles the length; squaring ρ_n doubles the inverse temperature:



A double leg denotes the n-site physical space. Taking traces gives

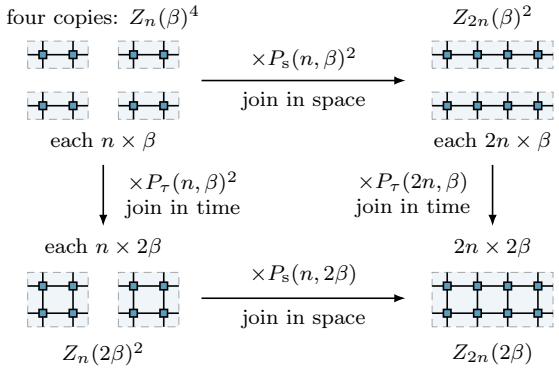
$$P_{\mathrm{s}}(n,\beta) = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2}, \qquad P_\tau(n,\beta) = \frac{Z_n(2\beta)}{Z_n(\beta)^2}. \tag{7}$$

Both operations replace normalized eigenvalues p_i by p_i^2/P. To bound the new purity P', use $1-P = 2\sum_{i<k} p_i p_k$:

$$1-P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \le \frac{(1-P)^2}{2P^2} =: f(1-P). \quad (8)$$

Dropping cross terms from $(\sum_{i<k} p_i p_k)^2$ gives the inequality. Here f increases on $[0,1)$, with $f(x) \sim x^2/2$ for small x.

Joining copies in space and time.— Squaring controls P_s when n doubles and P_τ when β doubles. To double both variables, we also need to relate P_s and P_τ. Rejoin the trace loops of four copies of $Z_n(\beta)$ in either order. Each rectangle is periodic; its width and height denote length and inverse temperature. Tensors are schematic; arrows give ratios of new to old contraction values.



Both orders give $Z_{2n}(2\beta)$. Equating their ratios, then using the same identity at $2n$, gives

$$P_\mathrm{s}(n,2\beta) = \frac{P_\mathrm{s}(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \quad (9)$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_\mathrm{s}(2n,2\beta)}{P_\mathrm{s}(2n,\beta)^2}. \quad (10)$$

Doubling estimate.— We now prove that, for $0 \le \varepsilon \le 1/100$,

$$\begin{aligned} & P_\mathrm{s}(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad & P_\mathrm{s}(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \quad (11)$$

The lengths n and $2n$ let the identities above reproduce this pair of purities at the next scale.

First bound the new transfer purity. Since $P_\tau(n,\beta) \le 1$, Eq. (9) gives $P_\mathrm{s}(n,2\beta) \ge (1-\varepsilon)^3$. Spatial blocking and Eq. (8) therefore give $P_\mathrm{s}(2n,2\beta) \ge 1-\delta$, where

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3}-1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \quad (12)$$

Here we used $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$.

Now bound the new Gibbs purity. Inserting the transfer bound into Eq. (10), with $\delta \le \varepsilon$, gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^3$. Squaring the Gibbs operator then gives $P_\tau(4n,2\beta) \ge 1-\delta$ by Eq. (8). This proves both bounds in Eq. (11).

Finite input and iteration.— We start the iteration from Proposition 5.5 of Ref. [1]:

$$P_\mathrm{s}(2304,784),\ P_\tau(4608,784) \ge 0.99. \quad (13)$$

The energy shift used there cancels from the purity ratios. After $j \ge 0$ doublings, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \quad (14)$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, Eq. (11) gives $P_\mathrm{s}(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$. The error $\varepsilon_j = \frac{1}{5}\exp[-\beta_j(\log 20)/784]$ decays exponentially in inverse temperature, as needed in Eq. (2).

These estimates hold at the doubled lengths. To cover every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$. Monotonicity of the ℓ^p norms in Eq. (4) gives

$$\begin{aligned} Z_L(\beta_j) &\le Z_{n_j}(\beta_j)^{L/n_j}, \\ Z_L(2\beta_j) &\ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \end{aligned} \quad (15)$$

Substitution into the Gibbs-purity ratio gives

$$\begin{aligned} P_\tau(L,\beta_j) &\ge \big[P_\tau(2n_j,\beta_j) P_\mathrm{s}(n_j,\beta_j)^2\big]^{L/(2n_j)} \\ &\ge (1-\varepsilon_j)^3 \ge 1-3\varepsilon_j > \tfrac{1}{2}. \end{aligned} \quad (16)$$

Here $L/(2n_j) \le 1$, and both purities in brackets are at least $1-\varepsilon_j$.

Recovering the gap.— We can now apply Eq. (2). Order the eigenvalues of $\rho_L(\beta_j)$ as $w_0 \ge w_1 \ge \cdots$. By Eqs. (5) and (16), $w_0 \ge 1-3\varepsilon_j > 1/2$. Two ground states would have equal largest weights, so the ground state is unique. The remaining weight bounds w_1, and hence

$$e^{-\beta_j \Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} < 20^{-2^j}. \quad (17)$$

The last inequality uses $3\varepsilon_j/(1-3\varepsilon_j) \le (60/97)20^{-2^j}$. Taking logarithms and dividing by $\beta_j = 784\,2^j$ gives, for every even $L \ge 2304$,

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821.} \quad (18)$$

Hence $\liminf_{L\to\infty,\ L\,\mathrm{even}} \Delta_L \ge (\log 20)/784 > 0$, conditional on Eq. (13).

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[3] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).